CODE 8 — ON ENERGY AND WATER

Version: V1.4
Declared alignment: Constitution V.29; Code 1 V1.6; Code 2 V1.5; Code 3 V1.5; Code 4 V1.4; Code 5 V1.4; Code 6 V1.5; Code 7 V1.7; C.1-Annex A V1.4; Code 9 V3.1; Code 10 V1.4; Code 11 V1.3; Code 12 V1.3; Code 13 V1.4; Code 14 V1.2; Code 15 V1.4; Code 16 V1.4.

Art. 1 — Subject matter and legal nature

1. This Code governs energy as an inalienable infrastructural commons pursuant to Article 3.1 of the Constitution and as a material condition for the continuity of the Basket, of health, of the Network and of production.
2. For the purposes of this Code, **energy** means electrical and thermal energy, cooling, motive power, storage systems and the other energy carriers necessary for the functioning of the Circles, of the Storehouses and of the CBO activities of Lists A and B.
3. This Code pursues the following aims:
   a) to guarantee federated energy autonomy and sobriety in consumption;
   b) to ensure access to the Energy Basket for vital needs, irrespective of competences or of the CBO performed;
   c) to prevent rents, speculative accumulation and critical proprietary dependencies;
   d) to guarantee the safety of installations and the progressive decarbonisation of the energy system.
4. Essential energy is not a commodity. Production, distribution and storage intended for vital needs may not be used to generate extractive profit or rankings of merit.
5. The management of energy is coordinated with Codes 1, 6, 7, 9, 10, 12, 13, 14 and 15, each for its respective competences.
6. This Code also governs the water cycle, water being a commons pursuant to Article 3.1 of the Constitution: abstraction, treatment to drinking-water standard, distribution, collection and purification. The agricultural use of water is governed by Code 9 and the siting of water infrastructure by Code 10.

Art. 2 — Principles

1. **Energy sovereignty.** Each Circle 1 pursues local self-production from renewable sources, accompanied by short-duration storage and a microgrid. Federated interconnections are oriented towards mutual aid and shall not create a structural dependency.
2. **Sobriety first.** The reduction of requirements precedes the increase of production. Every CBO Plan shall include an energy balance and the applicable efficiency measures.
3. **Priority of use and management of scarcity.** The hierarchy provided for in Article 32 of Code 1 applies:
   a) level 1: vital emergencies and the guaranteed minimum share of the Energy Basket;
   b) level 2: critical infrastructure for health, the Network, the Health Storehouse, servers and the conservation of the Basket;
   c) level 3: safety and maintenance;
   d) level 4: CBO of List A;
   e) level 5: other uses.
4. Level 1 constitutes a minimum threshold of continuity and may not be reduced, except in the case of material impossibility or documented extreme emergency. In the event of a collective health emergency, Article 2-bis of Code 6 applies for the relevant aspects; in the other cases the urgency procedures of Code 3 apply.

5. In the event of scarcity, the Assembly shall resolve on the reductions on a proposal from the Energy and Water Council, applying them progressively from level 5 to the higher levels. Level 1 may be reduced only in the case of temporary technical impossibility, with immediate activation of the federated mutualism procedure and communication to the Guarantee Circle.
6. **Non-commodification of knowledge.** It is prohibited to patent the knowledge essential for local renewable production or to impose paywalls on life-saving energy STVs.
7. **Transparency.** Every installation has an STV and a Safety Sheet. The balance of production and consumption is tracked on the Network in aggregate and non-profiling form in accordance with Article 6 and Code 7.
8. **Repairability and maintenance.** Installations are designed and managed, as far as technically possible, so as to allow local L1 or L2 maintenance, repair and reuse of components and access to the necessary documentation.

Art. 3 — Energy and Water Council

1. Each Circle 1 shall establish an Energy and Water Council composed of three to five members with L2 or L3 competences in energy, electrical engineering, thermal engineering, hydraulics, water quality or occupational safety.
2. The members are drawn by lot in accordance with Article 12-bis of Code 3, remain in office for one year and may not be drawn by lot again for the same Council before twelve months have elapsed from the end of the term of office.
3. The Energy and Water Council:

   a) manages the Energy Storehouse and the Energy Basket;

   b) draws up and updates the Safety Sheets, as regards the energy aspects;

   c) proposes and updates the STVs of the installations in accordance with Code 2;

   d) manages the energy balance of the Circle and the microgrid;

   e) organises L1 training in electrical safety;

   f) feeds the Dependency Register provided for in Code 15;

   g) coordinates its activity with the Network Council provided for in Code 7 and with the Health Council provided for in Code 6.
4. The Energy and Water Council has no allocative powers over goods outside the energy sector, nor disciplinary powers. It may not access individual health data.
5. The Energy and Water Council may not moderate, remove or modify deliberative or political content for reasons of merit, orientation or dissent. It may intervene without delay only to prevent an immediate technical risk to the security of the Network or of the installation; the intervention shall be limited to what is necessary, documented and subject to subsequent verification.
6. The acts of the Council are divided into:

   a) **bound technical acts:** execution of an STV or of a Safety Sheet already in force. They are documented and published in accordance with the applicable levels of access, without the need for a new resolution;

   b) **discretionary proposals and decisions:** these are adopted in accordance with the consensus method provided for in Code 3, after publication of the proposal, opening of a period of not less than seventy-two hours for objections and recording of the outcome in the minutes.
7. Silence may be regarded as absence of objections only if the proposal has been published and the prescribed period has elapsed. Failure to publish, or the absence of a time limit, does not constitute consensus.

Art. 4 — Energy Basket and determination of requirements

1. The Energy Basket is part of the Sufficiency Basket provided for in Code 1. It comprises the guaranteed minimum share necessary to:

    a) preserve foodstuffs;

    b) cook food;

    c) have domestic hot water;

    d) guarantee basic lighting;

    e) recharge essential devices;

    f) ensure minimum winter heating and the cooling necessary to protect health;

    g) keep the Health Storehouse and the Network node operational.

2. Access to the Energy Basket may not be made conditional on the CBO, on the possession of competences or on physical presence in a specific place.

3. The unavailability of the minimum share necessary for a vital need constitutes a level 1 emergency.

4. The guaranteed minimum share is determined by the Assembly on a proposal from the Energy and Water Council and with the opinion of the Health Council for the aspects concerning frail persons. Without prejudice to all the services of paragraph 1, on first application domestic hot water is not less than twenty litres per person per day and minimum winter heating ensures in the inhabited rooms a temperature of not less than eighteen degrees Celsius. On a proposal from the Health Council, the Assembly may set a higher value for age groups. A temperature that the Assembly, on a proposal from the Health Council, sets as dangerous to health constitutes an emergency within the meaning of paragraph 3; until that resolution, the value indicated by the Health Council applies. Article 2, paragraph 2, applies: reducing the requirement precedes increasing production.

5. The determination shall consider at least:

    a) the climatic conditions;

    b) the characteristics of the buildings;

    c) the needs of frail persons in accordance with Code 12;

    d) the conditions relevant to health safety in accordance with Code 6;

    e) the technical availability and the characteristics of the microgrid.

6. The share and the calculation criteria shall be reviewed at least annually and whenever a significant climatic, health or infrastructural change occurs. They are documented in the Energy Basket STV and published in accordance with the applicable levels of access.

7. The determination may not be based on individual profiling of consumption. Any data necessary to guarantee a health need shall be processed in accordance with Codes 6 and 7.

Art. 4-bis — Water Basket

1. The Water Basket is part of the Sufficiency Basket provided for in Code 1. It comprises the drinking water necessary for drinking, cooking and providing for personal and domestic hygiene, as well as the water necessary for health and care services.

2. Access to the Water Basket may not be made conditional on the CBO, on the possession of competences or on physical presence in a specific place. Drinking water is guaranteed to anyone who is in the territory held in custody by the Circle, pursuant to Articles 1.3 and 4.9 of the Constitution.

3. The guaranteed minimum share is determined by the Assembly on a proposal from the Energy and Water Council, with the opinion of the Health Council. On first application it shall not be less than fifty litres per person per day.

4. The unavailability of the minimum share constitutes a level 1 emergency.

5. In the event of scarcity, drinking water for human use and the water necessary for health and care

services take precedence over every other use. For agricultural uses the priorities of Article 10, paragraph 8, of Code 9 apply.

Art. 5 — Production, STV and energy sources

1. Every installation has an STV in force indicating at least:
   a) the energy source;
   b) the power and the efficiency;
   c) the materials and components;
   d) the expected life cycle;
   e) the maintenance and the competences required;
   f) the arrangements for disposal or recovery;
   g) the risks, the operating limits and the safety measures;
   h) the shutdown and restart procedure;
   i) the technological and logistical dependencies.
2. Preference is given to solar sources, small-scale wind, small-scale hydroelectric where compatible with the habitat, low-enthalpy geothermal, residual biomass not in competition with food production, and heat recovery.
3. Fossil sources may be used only on a transitional basis for vital functions of level 1 or 2, when a renewable source is not immediately available. The use shall be recorded in the Dependency Register and accompanied by an Exit Plan with a person responsible and a fixed deadline.
4. The construction of new installations powered by non-renewable sources is prohibited, pursuant to Article 15.4 of the Constitution. Existing installations shall be decommissioned in accordance with the Exit Plans.
5. The transitional use of existing fossil installations for uses of level 3, 4 or 5 is permitted only by resolution of the General Assembly by qualified majority and shall provide cumulatively for:
   a) the documented technical reasons for the absence of a practicable alternative;
   b) a duration not exceeding twelve months;
   c) compensatory measures;
   d) a binding Substitution Plan;
   e) a review at least every six months.
6. The transitional use may not be renewed beyond twenty-four months in total, except in the case of an extreme emergency documented and resolved upon in accordance with the urgency procedures of Code 3. Article 2-bis of Code 6 applies only where the emergency is of a collective health nature.
7. Installations shall favour, where possible, self-construction or local maintenance by persons with validated L2 competence, the use of repairable components and open hardware.

Art. 6 — Microgrid, storage, distribution and metering

1. Each Circle 1 manages a local microgrid that can be isolated in sections, equipped with overload protection and with a human shutdown control accessible in accordance with the Safety Sheet.
2. The metering of consumption is permitted exclusively for the technical purposes of safety, energy balance, maintenance and verification of the Basket.
3. The following measurements are permitted on an ordinary basis:
   a) aggregate data per installation;
   b) aggregate data per building;
   c) aggregate data per microgrid.

4. High-frequency individual metering aimed at the profiling of behaviour or of personal life is prohibited.
5. Metering at the level of the dwelling unit is permitted only where it is necessary to guarantee the Basket or safety and provided that:

   a) the data are not published at individual level;

   b) the data are aggregated over a period not shorter than that strictly necessary;

   c) technical measures are adopted to prevent re-identification;

   d) the case of a dwelling occupied by a single person is handled in such a way that the data cannot be traced back to that person's personal life;

   e) the rules on minimisation, access and retention of Code 7, Article 2.2, are applied.
6. The granularity, the retention periods and the persons authorised to have access are defined in the STV. The purpose and the necessity of the metering shall be reviewed periodically.
7. Short-duration storage may include stationary LFP batteries or equivalent, thermal storage and hydraulic storage, according to technical and environmental compatibility.
8. Storage capacity is sized according to the Basket and the requirements of levels 1–3. It may not be aimed at speculation or individual accumulation to the detriment of common availability.
9. No Circle may depend structurally on a single other Circle for the supply of the Energy Basket.
10. Interconnections between microgrids are used for mutual aid in emergencies. Every interconnection shall be accompanied by an electrical Safety Sheet.

Art. 6-bis — Water cycle

1. Each Circle 1 manages, as far as possible locally, abstraction, rainwater harvesting, storage, treatment to drinking-water standard, distribution, the collection and purification of wastewater, and reuse. Every installation has an STV and a Safety Sheet.
2. The quality of drinking water is checked periodically in accordance with the Drinking Water STV, adopted with the binding opinion of the Health Council. The results shall be published. In the event of non-conformity, distribution shall be suspended without delay and an alternative source shall be activated.
3. Withdrawals shall remain within the recharge capacity of the water bodies and shall guarantee the ecological flow, pursuant to Article 14.6 of the Constitution. New abstractions require the mandatory opinion of the Biodiversity Council and of the Habitat and Territory Council.
4. Wastewater shall be purified and, where possible, reused. The reuse of water and of purification residues is permitted only if it complies with the applicable STV and with the opinion of the Health Council.
5. The metering of water consumption follows the rules of Article 6, paragraphs 2 to 6. Individual profiling is prohibited.
6. Where technically possible, two independent sources and a reserve equal to at least seventy-two hours of the Water Basket shall be provided, consistently with Article 39 of Code 1.
7. Where several Circles depend on the same river basin, withdrawals shall be coordinated in the competent Circle 2 or Circle 3, with a Dossier. No Circle may deprive the Circles downstream of the water necessary for their Basket.
8. Water dependencies, including connection to external networks during the transition, shall be entered in the Dependency Register in accordance with Article 8.

Art. 7 — Safety, PPE and training

1. No activity relating to the production, transformation, distribution, maintenance or shutdown of energy may be carried out without a Safety Sheet that is in force and adequate.

2. The Energy and Water Council validates the technical aspects of the installation, the electrical risks and the isolation and shutdown procedures. The Health Council gives the mandatory opinion on the health aspects, on the PPE within its competence and on the consequences for health. The competences are exercised separately and coordinated in the same Sheet or in linked documents.
3. Electrical PPE is provided free of charge and managed as a commons in accordance with the coordination between the competent Storehouses, Code 6 and Code 14. It shall be maintained and tracked. The absence of a necessary item of PPE entails the suspension of the activity concerned and does not constitute a breach attributable to the person to whom the PPE was not provided.
4. L1 training in electrical safety is mandatory for access to the installations under the conditions defined by the Safety Sheet.
5. L2 competence is required for installation and specialist maintenance; L3 competence is required for the design and validation of the STVs.
6. Anyone who detects a serious and immediate electrical risk may interrupt the activity and report it, and may not suffer any detriment for having acted in good faith. The procedures of Code 14 apply.

Art. 8 — Register of energy Dependencies

1. The Energy and Water Council feeds the Dependency Register provided for in Code 15, distinguishing at least:

   a) **energy dependencies:** external sources, transitional fossil fuels or external networks necessary for the Basket;

   b) **technological dependencies:** proprietary inverters, closed BMS systems, closed software, imported critical components or unverifiable firmware;

   c) **logistical dependencies:** materials or components not available within the Confederation or not repairable locally;

   d) **water dependencies:** external sources, networks or treatments necessary for the Water Basket.
2. Every critical dependency shall be accompanied by an Exit Plan indicating:

   a) the objective;

   b) the person responsible;

   c) the fixed final deadline;

   d) the continuity measures;

   e) the risk-reduction measures during the transition.
3. Priority is given to the plans relating to dependencies that affect the Energy Basket or functions of levels 1 and 2.
4. Energy autonomy shall be subject to a federated audit at least annually by independent persons with L3 competence, selected by sortition. The outcome shall be recorded and published in a form compatible with the security of infrastructure.

Art. 9 — Prohibitions and guarantees

1. It is prohibited to:

   a) generate rent from the essential energy of the Basket;

   b) impose paywalls on life-saving energy STVs;

   c) patent the essential knowledge necessary for local self-production;

   d) accumulate energy individually beyond the technical limits resolved upon by the Assembly, where the accumulation compromises the availability of the Basket or the balance of the microgrid;

   e) interrupt the Energy Basket or the Water Basket as a sanction;

   f) cede functional control of the microgrid to a party external to the Confederation.

2. Breaches of paragraph 1 constitute a serious procedural defect pursuant to Code 4. The suspension or deactivation of the non-compliant function shall be proportionate and accompanied, where technically possible, by the continuity of levels 1–3 in accordance with the Safety Sheet.

3. Total shutdown shall be ordered only where necessary to prevent further harm. For critical or vital functions, a continuity procedure and a compliant substitute function shall be activated, where technically possible.

4. An appeal against the decision may be lodged with the Guarantee Circle within fourteen days on grounds of procedural defect, manifest technical unreasonableness, discrimination or disproportion. Code 4 applies.

Art. 10 — Transitional provisions and linkages

1. This Code is coordinated with:

   a) Code 1, Articles 18 and 19 for the Basket and scarcity and Article 32 for the hierarchy of priorities;

   b) Code 2, for the STVs;

   c) Code 3, for resolutions and the urgency procedures;

   d) Code 4, for Guarantee and appeals;

   e) Code 5, for the validation of L2 and L3 competences in energy matters;

   f) Code 6, for PPE, health and health safety;

   g) Code 7, for servers, the Network and non-surveillance pursuant to Article 2.2;

   h) Code 12, for the protection of frail persons;

   i) Code 14, for general safety;

   j) Code 15, for the Dependency Register.

2. Within six months of the entry into force, each Circle 1 shall:

   a) draw up the energy balance;

   b) take a census of the existing installations;

   c) prepare a provisional STV and a minimum Safety Sheet for installations that lack them;

   d) guarantee the minimum Energy Basket, including by means of a safe transitional source.

3. Within twelve months, each Circle 1 shall activate a minimum microgrid that can be isolated in sections, short-duration storage for the requirements of levels 1–3 and a Network node with autonomous or emergency power supply.

4. Within twenty-four months an Exit Plan from fossil sources shall be prepared for the uses of levels 4 and 5 and for critical dependencies. Every plan shall indicate a fixed final deadline.

5. Anyone who manages energy installations on the date of entry into force may retain a provisional validation in order to guarantee continuity:

   a) for up to six months for functions corresponding to L3;

   b) for up to twelve months for functions corresponding to L2.

The conditions of Article 10, paragraph 3, of Code 5 apply.

6. Existing functions or installations that breach Article 9 shall be brought into compliance or deactivated within ninety days.

7. If immediate deactivation compromises a vital function of level 1 or 2, the Assembly shall approve within the same period a Substitution Plan or a plan for progressive deactivation, with a fixed final deadline and continuity measures.